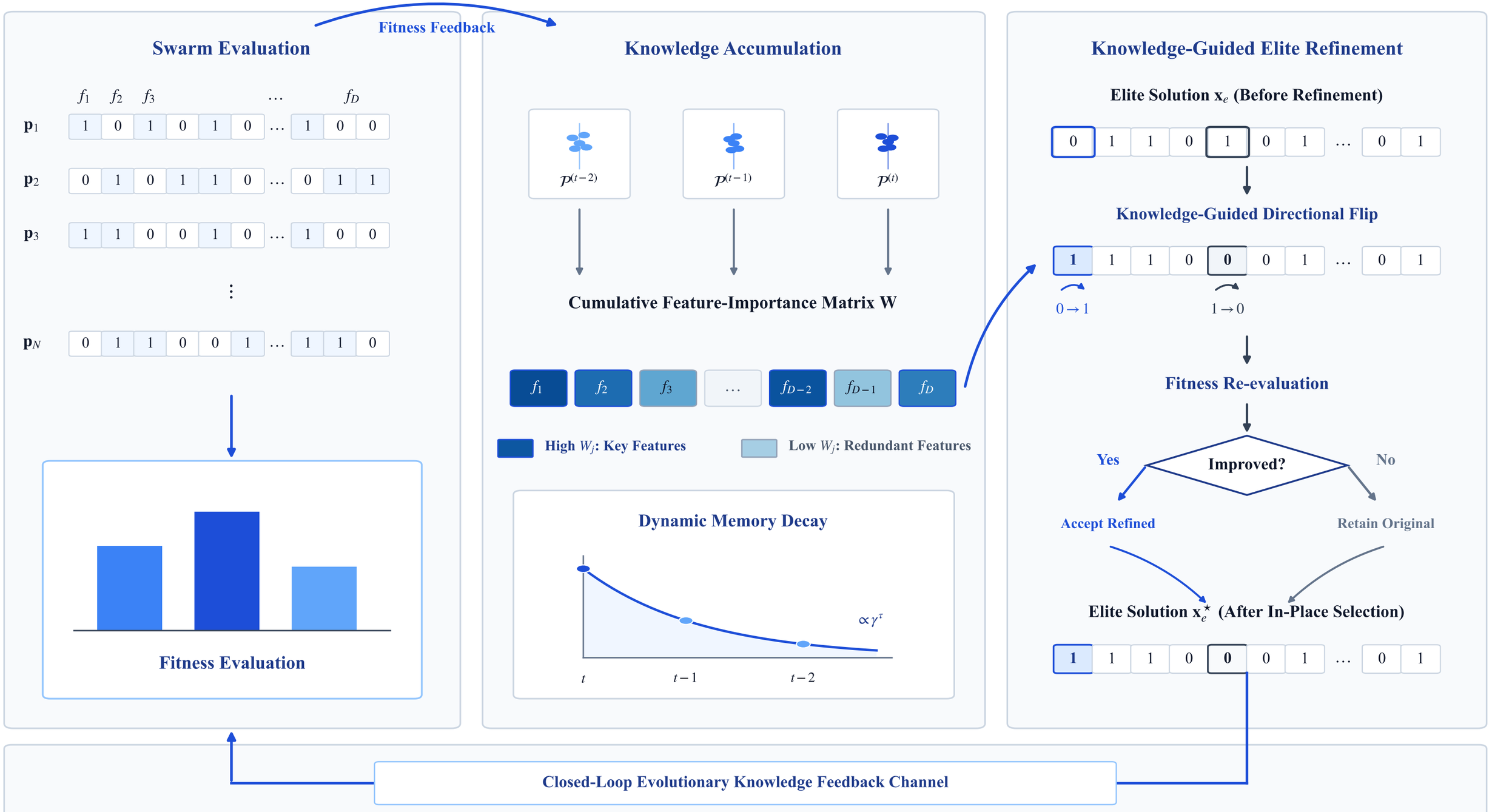

Fitness Feedback
Swarm Evaluation
Knowledge Accumulation
Knowledge-Guided Elite Refinement
Cumulative Feature-Importance Matrix W
Fitness Evaluation
High W_j: Key Features
Low W_j: Redundant Features
Dynamic Memory Decay
Elite Solution x_e (Before Refinement)
Knowledge-Guided Directional Flip
Fitness Re-evaluation
Improved?
Yes
No
Accept Refined
Retain Original
Elite Solution $x_e^\star$ (After In-Place Selection)
Closed-Loop Evolutionary Knowledge Feedback Channel